

Basics of FE-Analysis

BK10A6400

Lecture 3 : Rod examples and derivation of a rod element

Marko Matikainen

LUT University, Finland

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- Choose which one (Python or MATLAB) you are going to use in the course - not both!
- If you choose MATLAB for bonus task 1, then all other bonus tasks and coding parts of exercises need to be solved with MATLAB.
- MATLAB is easier (if you do not have much programming background), since almost all codes shown during the lectures are written in MATLAB.
- Do bonus tasks on time. Start as soon as possible so you will have time to ask...

RECAP: The principle of minimum potential energy for a rod (truss) element

Displacement field:

$$u_h = \mathbf{N} \bar{\mathbf{u}} \quad (1)$$

Strain (linear strain - displacement relationship):

$$\varepsilon = \frac{du_h}{dx} = \frac{d\mathbf{N}}{dx} \bar{\mathbf{u}} = \mathbf{B} \bar{\mathbf{u}} \quad (2)$$

where $\mathbf{B}$ is so-called kinematic matrix.

Constitutive equation (linear stress - strain relationship):

$$\sigma = E \varepsilon \quad (3)$$

Internal and external energies:

$$W_{int} = \frac{1}{2} EA \int_x \varepsilon^2 dx = \frac{1}{2} \bar{\mathbf{u}}^T \underbrace{EA \int_0^L \frac{\partial \mathbf{N}^T}{\partial x} \frac{\partial \mathbf{N}}{\partial x} dx}_{\bar{\mathbf{K}}} \bar{\mathbf{u}} \quad (4)$$

RECAP: The principle of minimum potential energy for a rod (truss) element

Internal energy is presented in the form of a stiffness matrix:

$$W_{int} = \frac{1}{2} \bar{\mathbf{u}}^T \underbrace{EA \int_0^L \frac{\partial \mathbf{N}^T}{\partial x} \frac{\partial \mathbf{N}}{\partial x} dx}_{\bar{\mathbf{K}}} \bar{\mathbf{u}} \quad (5)$$

External energy:

$$W_{ext} = \bar{\mathbf{f}}^T \bar{\mathbf{u}} \quad (6)$$

Minimizing energy:

$$\frac{\partial \Pi}{\partial \bar{\mathbf{u}}} = 0 \quad (7)$$

$$\bar{\mathbf{K}} \bar{\mathbf{u}} - \bar{\mathbf{f}} = 0 \quad (8)$$

RECAP: General derivation for a stiffness matrix of a rod element

Strain:

$$\varepsilon = \frac{du_h}{dx} \quad (9)$$

Constitutive equation:

$$\sigma = E\varepsilon \quad (10)$$

Strain energy:

$$W_{int} = \frac{1}{2} \int_V \sigma \varepsilon dV = \frac{1}{2} EA \int_x \varepsilon^2 dx \quad (11)$$

The vector of elemental internal forces:

$$\bar{\mathbf{f}}_{int} = \frac{\partial W_{int}}{\partial \bar{\mathbf{u}}} \quad (12)$$

Elemental local stiffness matrix:

$$\bar{\mathbf{K}} = \frac{\partial \bar{\mathbf{f}}_{int}}{\partial \bar{\mathbf{u}}} \quad (13)$$

RECAP: Displacement field approximation

$$\mathbf{u}_h = \mathbf{N}\mathbf{u} \quad (14)$$

- Shape functions describe shape of the displacement field.
- The shape function gets a value of 1 at that node and a value of zero at all other nodes.
- How to derive shape functions?

Example: Derive shape functions for a two-node rod elements in MATLAB

Displacement field:

$$u_h := a_0 + a_1 x = \mathbf{a}^T \mathbf{p} \quad (15)$$

Then giving boundary conditions $u_h(x=0) := u_1$ and $u_h(x=L) := u_2$:

$$\begin{cases} u_h(0) := a_0 = u_1 \\ u_h(L) := a_0 + a_1 L = u_2 \end{cases} \quad (16)$$

Eqs. (20) in a matrix form:

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 1 & L \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}}_{\mathbf{a}} = \underbrace{\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}}_{\mathbf{u}} \quad (17)$$

Example: Derive shape functions for a two-node rod element in MATLAB

By solving a_i 's:

$$\mathbf{a} = \mathbf{A}^{-1}\mathbf{u} \quad (18)$$

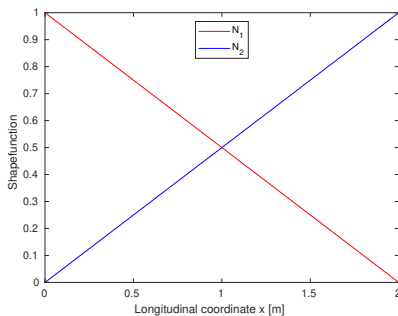
and substituting them into Eq.(15):

$$u_h = (\mathbf{A}^{-1}\mathbf{u})^T \mathbf{p} = \underbrace{\mathbf{p}^T \mathbf{A}^{-1}}_N \mathbf{u} \quad (19)$$

where N includes shape functions (known as ansatz functions in German literature).

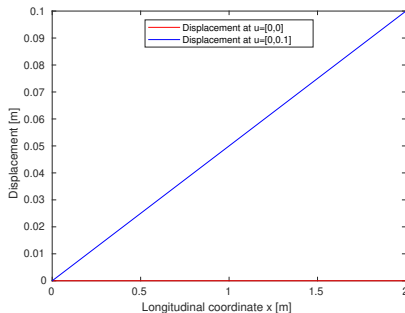
Example: Shape functions

- $N_1 = 1 - x/L$, $N_2 = x/L$
- The shape function gets a value of 1 at that node and a value of zero at all other nodes.
- $N_1(x = 0) = 1$, $N_1(x = L) = 0$
- $N_2(x = 0) = 0$, $N_2(x = L) = 1$



Example: displacement fields

- $u_h(x) = \mathbf{N}(x)\mathbf{u}$
- $\mathbf{N} = [N_1 \quad N_2]$ and $\mathbf{u} = [u_1 \quad u_2]^T$



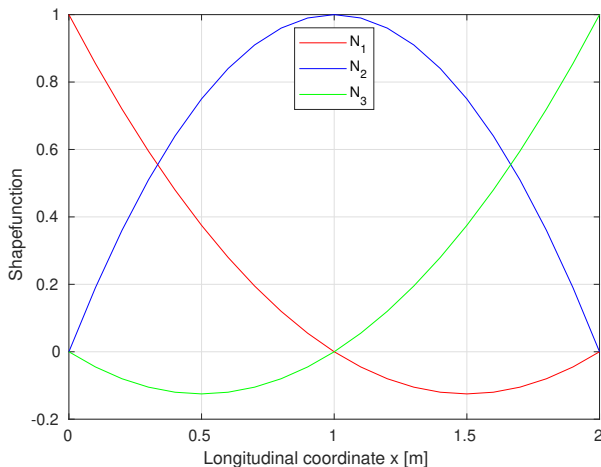
Example in a class: three-noded rod element

- Three nodal displacement coordinates u_1, u_2, u_3
- Need three polynomials $1, x, x^2$

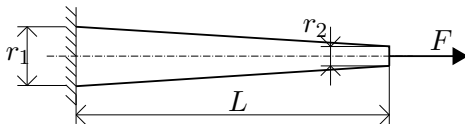
$$\begin{cases} u_h(0) := a_0 = u_1 \\ u_h(L/2) := a_0 + a_1 L/2 + \dots = u_2 \\ u_h(L) := a_0 + \dots = u_3 \end{cases} \quad . \quad (20)$$

- Find constants for a matrix A
- Solve shape functions $N = p^T A^{-1}$

Example in a class: Shape functions for a three-noded rod element



Example: Tapered rod structure



- Solve displacements at the free end.
- The length of a structure is 1000 mm
- The radius of the circular cross-section at the fixed end is 50 mm, and at the free end is 25 mm.
- Young's modulus is 210000 Mpa
- The applied force at the free end is 1000 N.
- Discretize the problem with three different meshes by using 2,4,8 rod elements and solve displacements.

Example: Analytical solution (known from previous courses)

Differential equation & boundary conditions

$$\left\{ \begin{array}{ll} \frac{d}{dx} \left(\frac{du}{dx} E A(x) \right) + q(x) = 0 & \text{Diff. eq.} \\ u(0) = 0 & \text{Essential bc} \\ E A(L) \frac{du}{dx} = F & \text{Natural bc} \end{array} \right. \quad (21)$$

Analytical solution (by hand):

$$u(L) = \frac{FL}{E(A_1 - A_0)} \ln \left(\frac{A_1}{A_0} \right) \quad (22)$$

Example: Analytical solution using MATLAB symbolic toolbox

```
% Introduce variables
syms u(x) A(x) E F L A0 A1

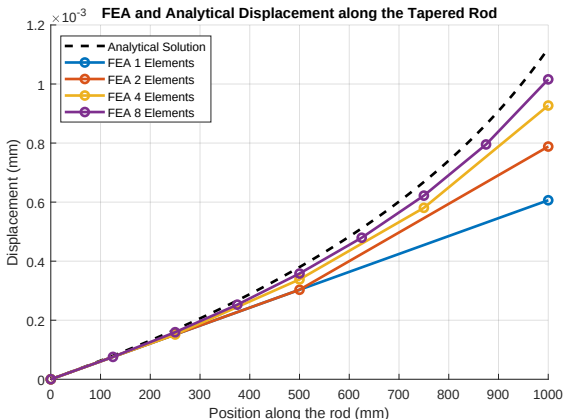
% Linearly tapered rod.
A(x) = A0*(1 - x/L) + A1*x/L;

epsilon = diff(u);
DE = -diff(E*A(x)*epsilon) == 0           % DE
bc = [u(0) == 0; E*A(L)*epsilon(L) == F] % BC

u = dsolve(DE,bc)
```

Example: Tapered rod structure: FE-solution

$$\overline{K} = \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad (23)$$



Bonus task 2: Connectivity

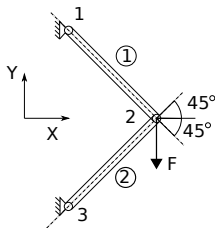
Sometimes, it is helpful to present elements' connectivity and corresponding nodal displacements in the table as follows:

	Node I	Node J	nodal displacements
Elem 1	1	2	u_1, u_2
Elem 2	2	3	u_2, u_3
Elem 3	3	4	u_3, u_4

Need to define a cross-sectional area for each element.

Example: Solve displacements at node 2 of a simple truss structure

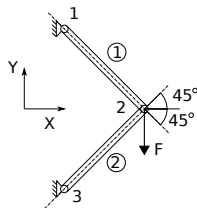
Solve displacements with rod finite elements.



- Global nodal displacements (u_X, u_Y) at each node.
- 3 nodes, 2 DOFs at node $\rightarrow$ a system of 6 DOF before taking into account the (linear) constraints that are applied by eliminating rows and columns of the stiffness matrix for a system.
- Constraints i.e. fixed DOF's:

$$u_{X_1} = 0, u_{Y_1} = 0, u_{X_3} = 0, u_{Y_3} = 0$$

Example: Solve displacements at node 2 of a simple truss structure



The parameters of the problem are given in Table.

	Young's modulus, E [MPa]	Cross section, A [mm ²]	Length, L [mm]
Truss 1	210000	100	1000
Truss 2	210000	50	1000
Force, F [N]	100 000		

Example: Simple truss structure problem: Connectivity

It is helpful to present elements' connectivity and corresponding nodal displacements in the table as follows:

	Node I	Node J	nodal displacements
Truss 1	1	2	u_1, v_1, u_2, v_2
Truss 2	2	3	u_2, v_2, u_3, v_3

It can be seen from the connectivity table what stiffness components of global elemental stiffness matrices $\mathbf{K}^{(1)}$ and $\mathbf{K}^{(2)}$ are needed to sum to mimic the structure correctly. After assembling, the global stiffness matrix should be a size of 6×6 .

Example: Simple truss structure problem

- Elem 1 consists of nodes 1 and 2
- Elem 2 consists of nodes 2 and 3
- Angle of a rod 1: -45 deg
- Angle of a rod 2: -135 deg

Transformation into a global coordinates system.

Rod 1:

$$\mathbf{K}_G^{(1)} = \mathbf{T}(\theta_1)^T \overline{\mathbf{K}} \mathbf{T}(\theta_1) \quad (24)$$

Rod 2:

$$\mathbf{K}_G^{(2)} = \mathbf{T}(\theta_2)^T \overline{\mathbf{K}} \mathbf{T}(\theta_2) \quad (25)$$

Rod 1:

$$\mathbf{K}_G^{(1)} = \mathbf{T}(\theta_1)^T \overline{\mathbf{K}} \mathbf{T}(\theta_1) \quad (26)$$

$$\mathbf{K}_G^{(1)} = \begin{matrix} & \begin{matrix} u_1 & v_1 & u_2 & v_2 \end{matrix} \\ \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} \\ k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} & k_{34}^{(1)} \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} & k_{44}^{(1)} \end{pmatrix} & \begin{matrix} u_1 \\ v_2 \\ u_2 \\ v_2 \end{matrix} \end{matrix}$$

Rod 2:

$$\mathbf{K}_G^{(2)} = \mathbf{T}(\theta_2)^T \overline{\mathbf{K}} \mathbf{T}(\theta_2) \quad (27)$$

$$\mathbf{K}_G^{(2)} = \begin{matrix} & \begin{matrix} u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{pmatrix} k_{11}^{(2)} & k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} \\ k_{21}^{(2)} & k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} \\ k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} \\ k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} \end{pmatrix} & \begin{matrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix} \end{matrix}$$

$$\mathbf{K}_G = \begin{matrix} & \begin{matrix} u_1 & v_1 & u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} & 0 & 0 \\ k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} & 0 & 0 \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} + k_{11}^{(2)} & k_{34}^{(1)} + k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} + k_{21}^{(2)} & k_{44}^{(1)} + k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} \\ 0 & 0 & k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} \\ 0 & 0 & k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} \end{pmatrix} & \begin{matrix} u_1 \\ v_1 \\ u_2 \\ u_2 \\ u_3 \\ v_3 \end{matrix} \end{matrix}$$

External forces are set up in a global coordinate system because our global stiffness matrix is also presented in a global coordinate system.

Force F is active at node 2 in the same direction with nodal displacement v_2 .

$$\mathbf{f}_G = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -F \\ 0 \\ 0 \end{pmatrix} \begin{matrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}$$

Boundary conditions

Fixed nodal displacements are $u_1 = v_1 = u_3 = v_3 = 0$. Let's eliminate corresponding rows and columns regarding FE-matrices.

$$\mathbf{f}_{G_{red}} = \begin{pmatrix} 0 \\ -F \end{pmatrix} \begin{matrix} u_2 \\ v_2 \end{matrix}$$

$$\mathbf{K}_{G_{red}} = \begin{matrix} & \begin{matrix} u_2 & v_2 \end{matrix} \\ \begin{pmatrix} k_{33}^{(1)} + k_{11}^{(2)} & k_{34}^{(1)} + k_{12}^{(2)} \\ k_{43}^{(1)} + k_{21}^{(2)} & k_{44}^{(1)} + k_{22}^{(2)} \end{pmatrix} & \begin{matrix} u_2 \\ v_2 \end{matrix} \end{matrix}$$

Solving global displacements

Note that $\mathbf{K}_{G_{red}}$ is symmetric i.e. $k_{ij} = k_{ji}$ (if not, there is a typo) and $\det \mathbf{K}_{G_{red}} > 0$ (if not, check BC's).

And then displacements (in a global coordinate system) can be solved as

$$\mathbf{u}_{G_{red}} = \mathbf{K}_{G_{red}}^{-1} \mathbf{f}_{G_{red}}$$

Elemental displacements and member forces

After solving $\mathbf{u}_{G_{red}}$, local elemental displacements (for an element i) can be computed as follows:

$$\bar{\mathbf{u}}^{(i)} = \mathbf{T}^{(i)} \mathbf{u}_G^{(i)} \quad (28)$$

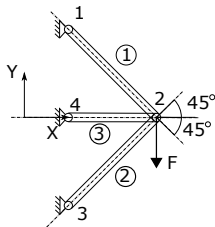
Member forces (for an element i) can be found as

$$\bar{\mathbf{f}}^{(i)} = \bar{\mathbf{K}}^{(i)} \bar{\mathbf{u}}^{(i)} \quad (29)$$

Note $\mathbf{u}_G^{(i)}$ includes element's i (solved nodal coordinates and nodal coordinates at supports).

Example: Solve displacements at node 2 of a simple truss structure of three rods

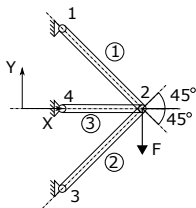
Solve displacements with rod finite elements.



- Global nodal displacements (u_X, u_Y) at each node.
- 4 nodes, 2 DOFs (global nodal displacements) at node $\rightarrow$ a system of 8 DOF before taking into account the (linear) constraints that are applied by eliminating rows and columns of the stiffness matrix for a system.
- Constraints i.e. fixed DOF's:

$$u_1 = 0, v_1 = 0, u_3 = 0, v_3 = 0, u_4 = 0, v_4 = 0$$

Bonus task: Solve displacements at node 2 of a simple truss structure of three rods



The parameters of the problem are given in Table.

	Young's modulus, E [MPa]	Cross section, A [mm ²]	Length, L [mm]
Truss 1	210000	100	1000
Truss 2	210000	50	1000
Truss 3	210000	$10 + 4n_{student}$	$1000 \sin(\pi/4)$
Force, F [N]	100 000		

Bonus task: simple truss structure with three rods

Connectivity

It is helpful to present the element connectivity and the corresponding nodal displacements in a table, as shown below:

Element	Node I	Node J	Nodal displacements
Truss 1	1	2	u_1, v_1, u_2, v_2
Truss 2	2	3	u_2, v_2, u_3, v_3
Truss 3	4	2	u_4, v_4, u_2, v_2

From the connectivity table, it can be seen which stiffness components of the global elemental stiffness matrices $\mathbf{K}_G^{(1)}$, $\mathbf{K}_G^{(2)}$, and $\mathbf{K}_G^{(3)}$ need to be assembled to correctly represent the structure. After assembly, the global stiffness matrix will have the size of 4×4 .

Bonus task: Simple truss structure problem

- Elem 1 consists of nodes 1 (start) and 2 (end)
- Elem 2 consists of nodes 2 (start) and 3 (end)
- Elem 3 consists of nodes 4 (start) and 2 (end)
- Angle of a rod 1: ?
- Angle of a rod 2: ?
- Angle of a rod 3: ?

Transformation into a global coordinates system.

Rod 1:

$$\mathbf{K}_G^{(1)} = \mathbf{T}(\theta_1)^T \bar{\mathbf{K}} \mathbf{T}(\theta_1) \quad (30)$$

Rod 2:

$$\mathbf{K}_G^{(2)} = \mathbf{T}(\theta_2)^T \bar{\mathbf{K}} \mathbf{T}(\theta_2) \quad (31)$$

Rod 3:

$$\mathbf{K}_G^{(3)} = \mathbf{T}(\theta_3)^T \bar{\mathbf{K}} \mathbf{T}(\theta_3) \quad (32)$$

Rod 1:

$$\mathbf{K}_G^{(1)} = \mathbf{T}(\theta_1)^T \overline{\mathbf{K}} \mathbf{T}(\theta_1) \quad (33)$$

$$\mathbf{K}_G^{(1)} = \begin{matrix} & \begin{matrix} u_1 & v_1 & u_2 & v_2 \end{matrix} \\ \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} \\ k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} & k_{34}^{(1)} \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} & k_{44}^{(1)} \end{pmatrix} & \begin{matrix} u_1 \\ v_2 \\ u_2 \\ v_2 \end{matrix} \end{matrix}$$

Rod 2:

$$\mathbf{K}_G^{(2)} = \mathbf{T}(\theta_2)^T \overline{\mathbf{K}} \mathbf{T}(\theta_2) \quad (34)$$

$$\mathbf{K}_G^{(2)} = \begin{matrix} & \begin{matrix} u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{pmatrix} k_{11}^{(2)} & k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} \\ k_{21}^{(2)} & k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} \\ k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} \\ k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} \end{pmatrix} & \begin{matrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix} \end{matrix}$$

Rod 3:

$$\mathbf{K}_G^{(3)} = \mathbf{T}(\theta_3)^T \overline{\mathbf{K}} \mathbf{T}(\theta_3) \quad (35)$$

$$\mathbf{K}_G^{(3)} = \begin{matrix} & \begin{matrix} u_4 & v_4 & u_2 & v_2 \end{matrix} \\ \begin{pmatrix} k_{11}^{(2)} & k_{12}^{(3)} & k_{13}^{(3)} & k_{14}^{(3)} \\ k_{21}^{(3)} & k_{22}^{(3)} & k_{23}^{(3)} & k_{24}^{(3)} \\ k_{31}^{(3)} & k_{32}^{(3)} & k_{33}^{(3)} & k_{34}^{(3)} \\ k_{41}^{(3)} & k_{42}^{(3)} & k_{43}^{(3)} & k_{44}^{(3)} \end{pmatrix} & \begin{matrix} u_4 \\ v_4 \\ u_2 \\ v_2 \end{matrix} \end{matrix}$$

Assembling

Note that all stiffnesses at common nodal displacement (global DOFs) in a structure need to be summed up.

$$\mathbf{K}_G = \begin{matrix} & \begin{matrix} u_1 & v_1 & & u_2 & & v_2 & & u_3 & v_3 & u_4 & v_4 \end{matrix} \\ \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & & k_{13}^{(1)} & & k_{14}^{(1)} & & 0 & 0 & 0 & 0 \\ k_{21}^{(1)} & k_{22}^{(1)} & & k_{23}^{(1)} & & k_{24}^{(1)} & & 0 & 0 & 0 & 0 \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} + k_{11}^{(2)} + k_{33}^{(3)} & k_{34}^{(1)} + k_{12}^{(2)} + k_{34}^{(3)} & k_{13}^{(2)} & k_{14}^{(2)} & k_{31}^{(3)} & k_{32}^{(3)} \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} + k_{21}^{(2)} + k_{43}^{(3)} & k_{44}^{(1)} + k_{22}^{(2)} + k_{44}^{(3)} & k_{23}^{(2)} & k_{24}^{(2)} & k_{41}^{(3)} & k_{42}^{(3)} \\ 0 & 0 & k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} & 0 & 0 \\ 0 & 0 & k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} & 0 & 0 \\ 0 & 0 & k_{13}^{(3)} & k_{14}^{(3)} & 0 & 0 & k_{11}^{(3)} & k_{12}^{(3)} \\ 0 & 0 & k_{23}^{(3)} & k_{24}^{(3)} & 0 & 0 & k_{21}^{(3)} & k_{22}^{(3)} \end{pmatrix} & \begin{matrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{matrix} \end{matrix}$$

External forces are set up in a global coordinate system because our global stiffness matrix is also presented in a global coordinate system.

Force F is active at node 2 in the same direction with nodal displacement v_2 .

$$\mathbf{f}_G = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -F \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{matrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{matrix}$$

Boundary conditions

Fixed nodal displacements are

$u_1 = v_1 = u_3 = v_3 = u_4 = v_4 = 0$. Let's eliminate corresponding rows and columns regarding FE-matrices.

$$\mathbf{f}_{G_{red}} = \begin{pmatrix} 0 \\ -F \end{pmatrix} \begin{matrix} u_2 \\ v_2 \end{matrix}$$

$$\mathbf{K}_{G_{red}} = \begin{matrix} & \begin{matrix} u_2 & v_2 \end{matrix} \\ \begin{pmatrix} k_{33}^{(1)} + k_{11}^{(2)} + k_{33}^{(3)} & k_{34}^{(1)} + k_{12}^{(2)} + k_{34}^{(3)} \\ k_{43}^{(1)} + k_{21}^{(2)} + k_{43}^{(3)} & k_{44}^{(1)} + k_{22}^{(2)} + k_{44}^{(3)} \end{pmatrix} & \begin{matrix} u_2 \\ u_2 \end{matrix} \end{matrix}$$

Solving global displacements

Note that $\mathbf{K}_{G_{red}}$ is symmetric i.e. $k_{ij} = k_{ji}$ (if not, there is a typo) and $\det \mathbf{K}_{G_{red}} > 0$ (if not, check BC's).

And then displacements (in a global coordinate system) can be solved as

$$\mathbf{u}_{G_{red}} = \mathbf{K}_{G_{red}}^{-1} \mathbf{f}_{G_{red}}$$

Elemental displacements and member forces

After solving $\mathbf{u}_{G_{red}}$, local elemental displacements (for an element i) can be computed as follows:

$$\bar{\mathbf{u}}^{(i)} = \mathbf{T}^{(i)} \mathbf{u}_G^{(i)} \quad (36)$$

Member forces (for an element i) can be found as

$$\bar{\mathbf{f}}^{(i)} = \bar{\mathbf{K}}^{(i)} \bar{\mathbf{u}}^{(i)} \quad (37)$$

Note $\mathbf{u}_G^{(i)}$ includes element's i (solved nodal coordinates and nodal coordinates at supports).